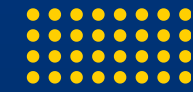


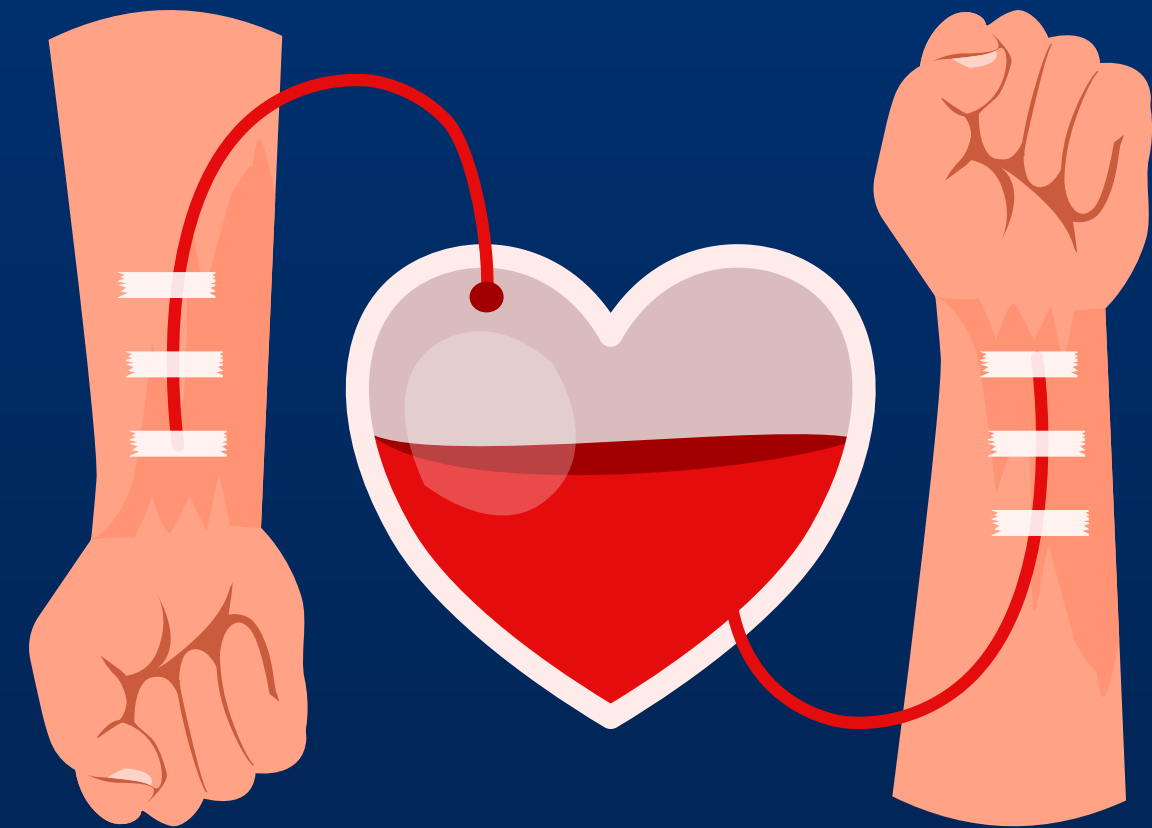
Regulation of CVS



Assist.Prof.Dr.Nifareeda Samerphob



Division of health and applied sciences,
Faculty of science, Prince of Songkla University

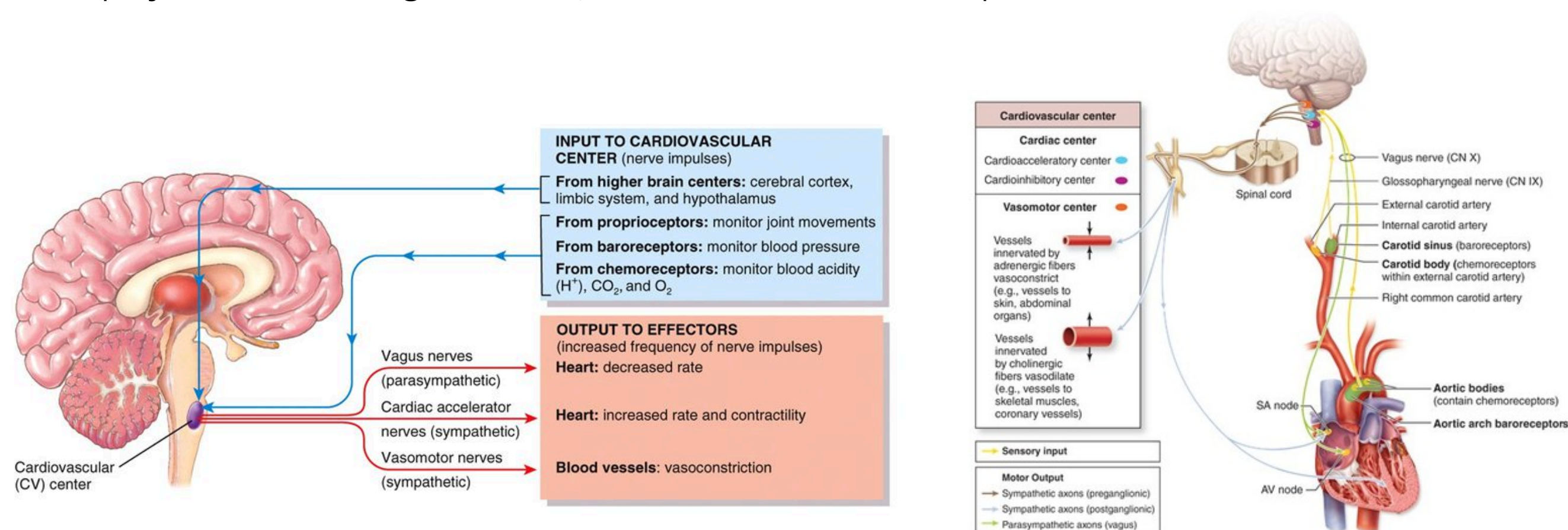


LEARNING OBJECTIVES

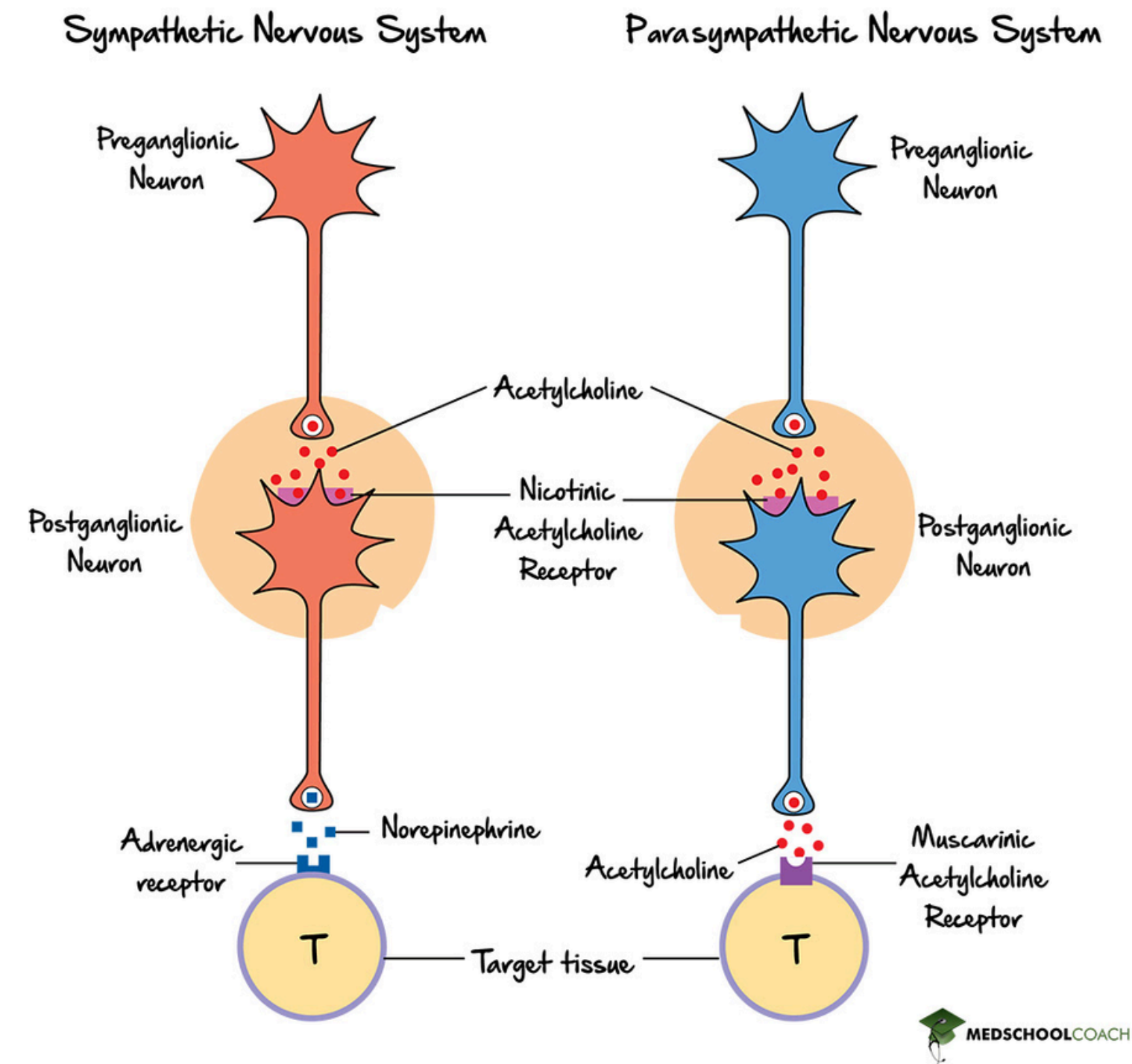
- 1. Cardiovascular center**
- 2. Intrinsic & extrinsic control of cardiac and blood vessels activity**
- 3. Control of arterial blood pressure**
 - 3.1 Short term mechanism**
 - 3.1.1 Baroreceptor reflex**
 - 3.1.2 Chemoreceptor reflex**
 - 3.2 Long term mechanism**
 - 3.2.1 Renin-angiotensin aldosterone system (RAAS)**
 - 3.2.2 Atrial natriuretic peptide (ANP)**
 - 3.2.3 Antidiuretic hormone (ADH)**

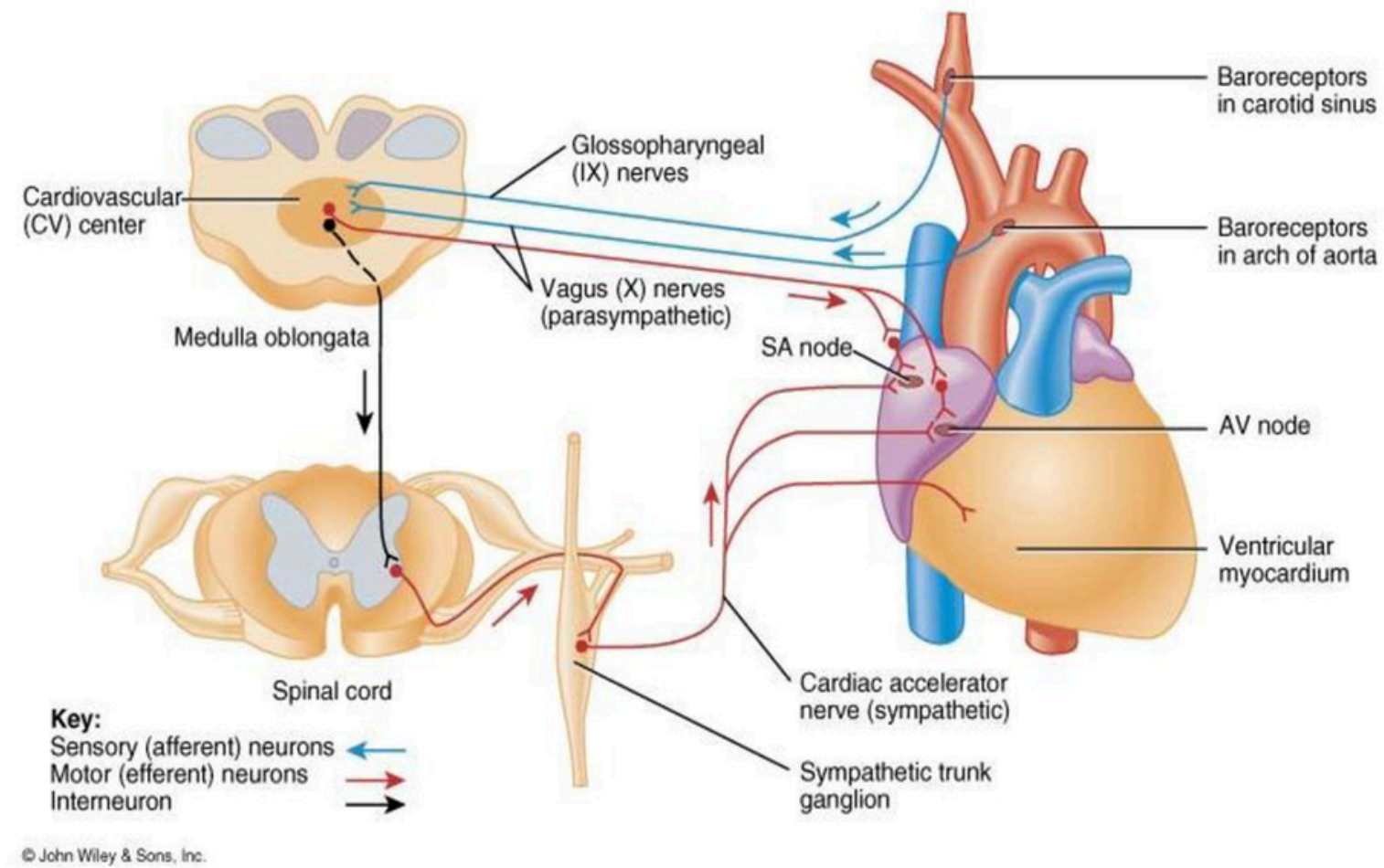
Cardiovascular center

- The cardiovascular center is a part of the human brain found in the **medulla oblongata**, consisting of
 - cardioaccelerator center,
 - cardioinhibitory center
 - vasomotor center..
- Numerous receptors in the circulatory system can:
 - detect changes in pH
 - detect stretch
 - detect chemical signal
- The centers play a role in altering heart rate, stroke volume and blood pressure to alter blood flow.



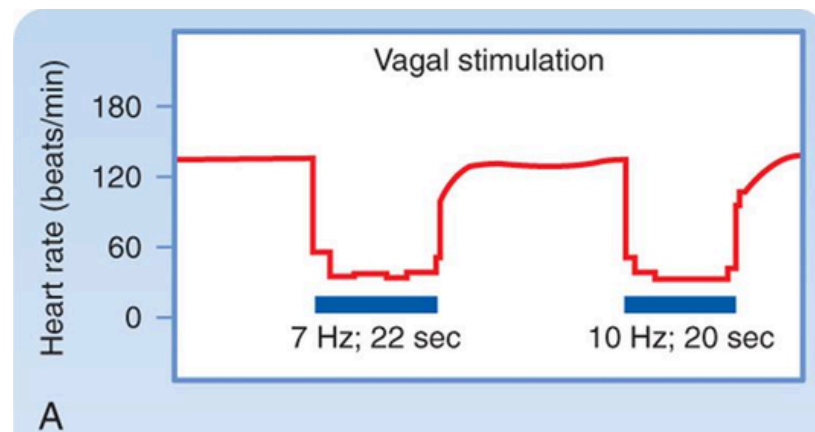
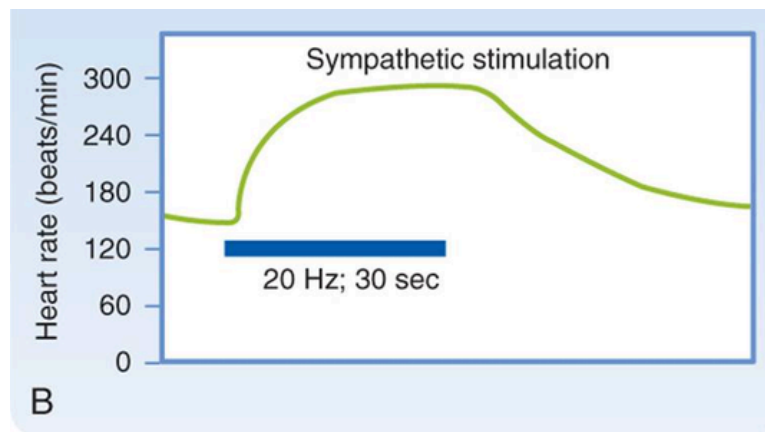
- The autonomic nervous system controls involuntary functions of visceral organs, including the heart and blood vessels.
- It operates through preganglionic fibers that originate in the spinal cord (sacral for parasympathetic, thoracic/lumbar for sympathetic) and synapse in autonomic ganglia on postganglionic fibers, which innervate target tissues.
- Parasympathetic division: Long preganglionic fibers synapse on short postganglionic fibers near the target tissue, causing localized effects. Acetylcholine (ACh) is released and binds to muscarinic receptors.
- Sympathetic division: Short preganglionic fibers synapse on long postganglionic fibers in paravertebral or collateral ganglia, causing diffuse effects. Postganglionic fibers release norepinephrine, which acts on α - or β -adrenergic receptors.
- Both divisions use acetylcholine as the neurotransmitter for preganglionic fibers, activating nicotinic receptors on postganglionic neurons.





Cardiac innervation:

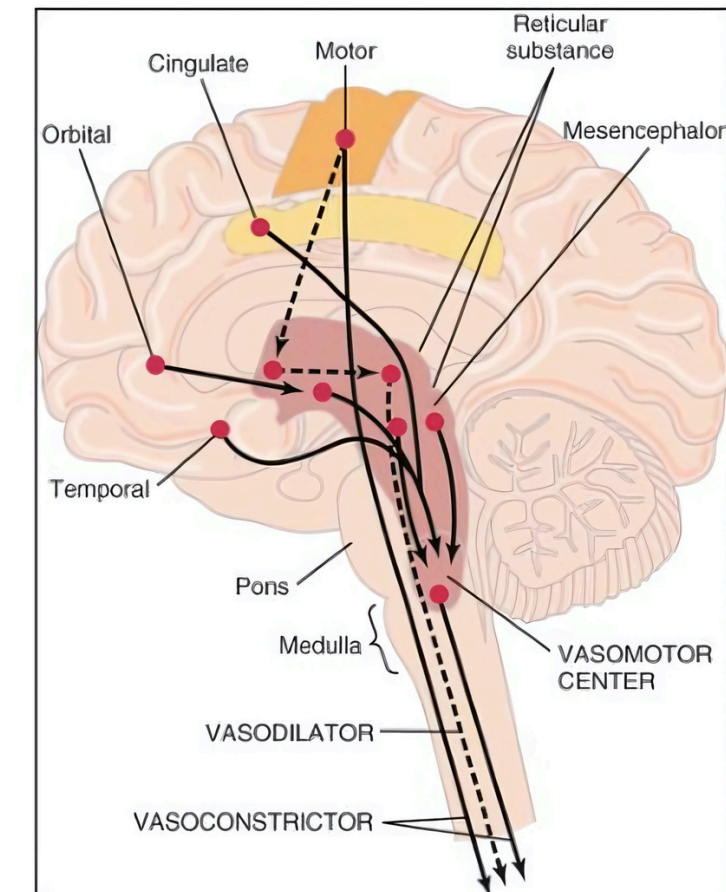
1. **Cardioaccelerator center:** increased sympathetic activity by releasing norepinephrine binding to beta1-adrenergic receptor at the cardiac membrane
 - 1.1 increased heart rate (positive chronotropic effect)
 - 1.2 increased conduction velocity (positive bromotropic effect)
 - 1.3 increased force of contraction (positive inotropic effect)
2. **Cardioinhibitory center:** increased vagal parasympathetic stimulation by releasing acetylcholine binding to muscarinic receptor at the cardiac membrane
 - 2.1 decreased heart rate (negative chronotropic effect)



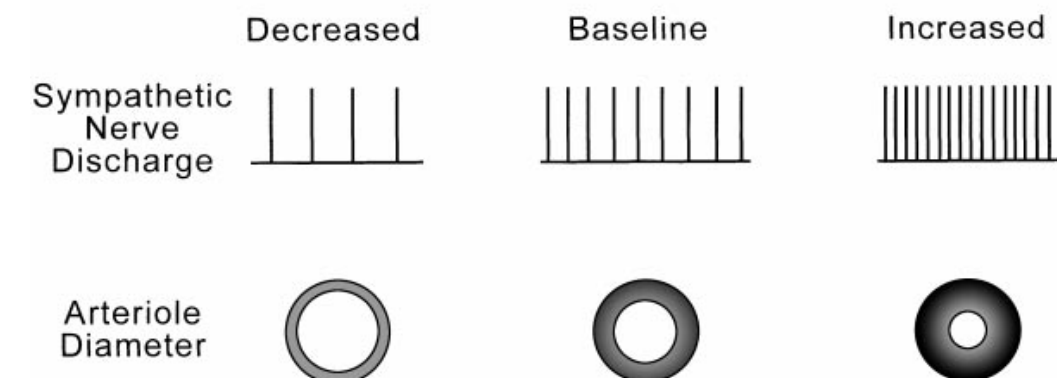
Koeppen & Stanton: Berne and Levy Physiology, 6th Edition.
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The **vasomotor center (VMC)**

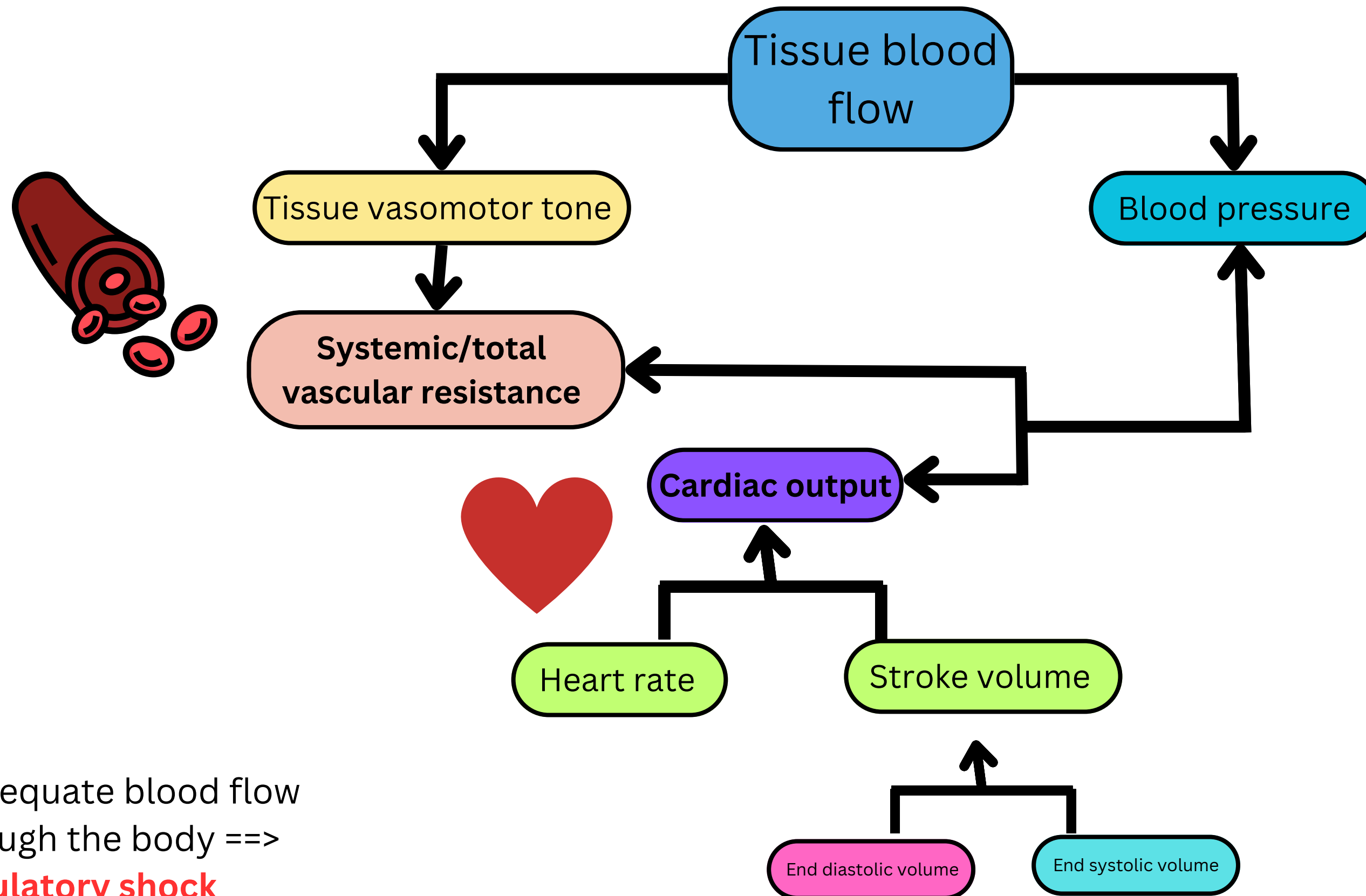
- VMC is located bilaterally in the reticular substance of the medulla and the lower third of the pons.
- The VMC transmits impulses downward through the cord to almost all blood vessels.
- The VMC is composed of:
 - **sensory area** (pons, medulla, hypothalamus, cerebral cortex, cingulate gyrus, amygdala, hippocampus)
 - **vasoconstrictor area** (sympathetic noradrenergic fibers)
 - **vasodilator area** (sympathetic cholinergic fibers)



Sympathetic Tone



The most important function of CVS is to **supply sufficient tissue blood flow**



$$F = P/R$$

$$ABP = CO \times TPR$$

$$CO = HR \times SV$$

$$SV = EDV - ESV$$

Inadequate blood flow
through the body ==>
circulatory shock

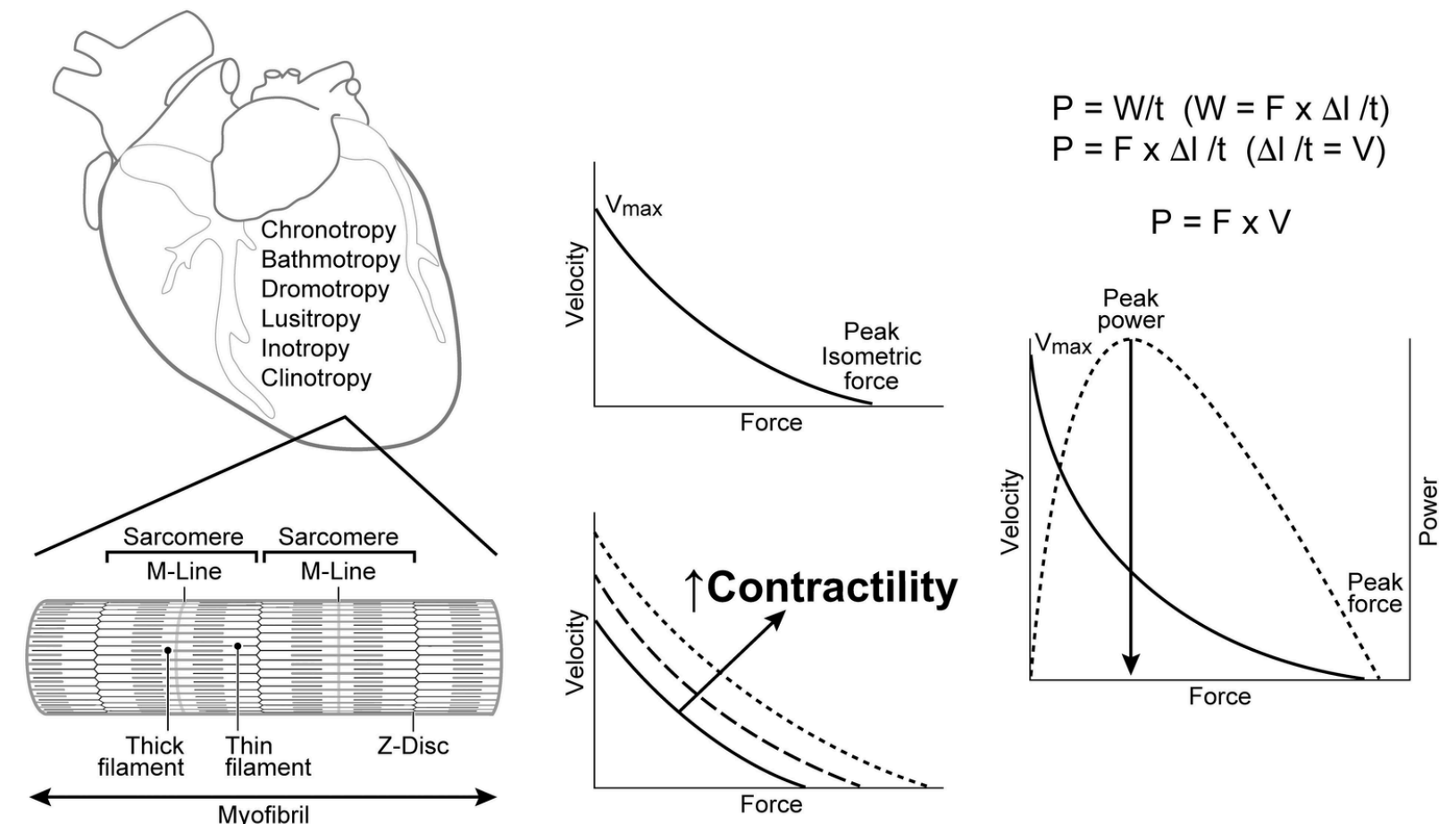
Intrinsic control in cardiac function (autoregulation)

Intrinsic control in cardiac function refers to the regulation of the heart's activity that originates from within the heart itself, without input from the nervous or endocrine systems.

$$\begin{array}{ccccc} \text{CO} & = & \text{SV} & \times & \text{HR} \\ (\text{mL/min}) & & (\text{mL/beat}) & & (\text{beats/min}) \end{array}$$

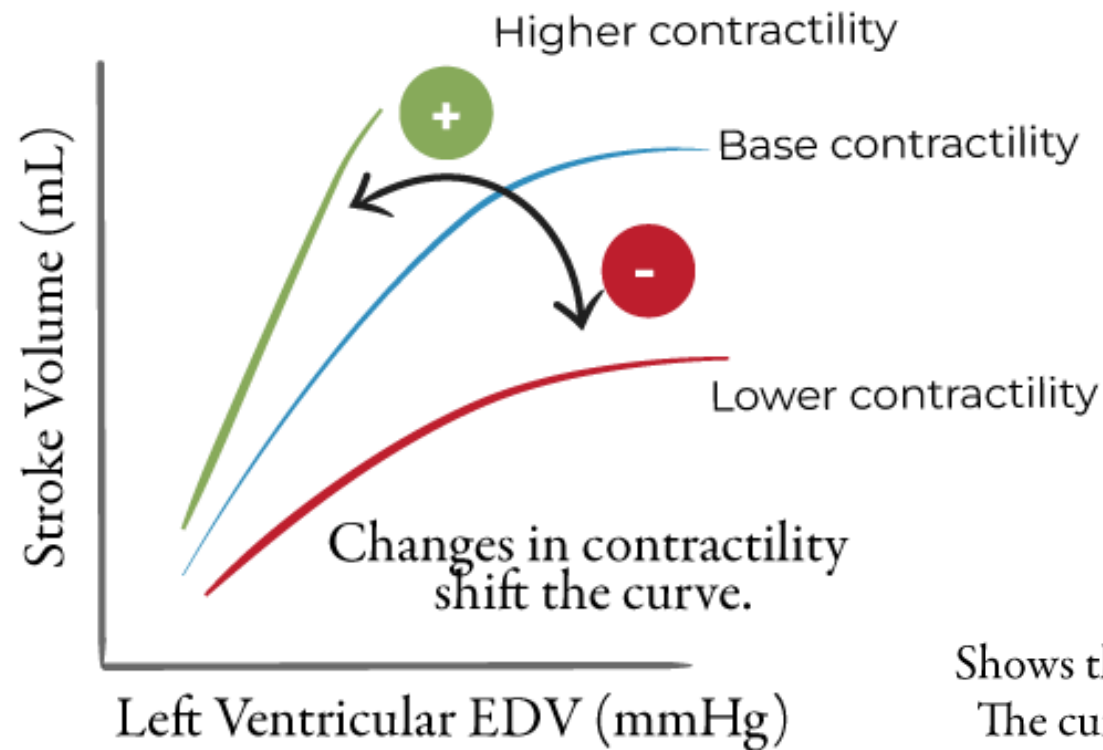
Intrinsic Control of Heartbeat

- Cardiac conduction system
- SA node
- AV node
- AV bundle
- Left and right bundle branches
- Purkinje fibers
- Allows impulse conduction to spread first to atria and then after delay to ventricles
- Allows for adequate filling of all chambers



V = Velocity; V_{max} = maximal velocity at no load; F = Force;
P = power; W = work; l = length; t = time; Δ = change

Regulation of the cardiac contraction



Frank-Starling Law

Shows that stroke volume changes as Preload/EDV does.
The curve shifts right/left with changes in contractility.

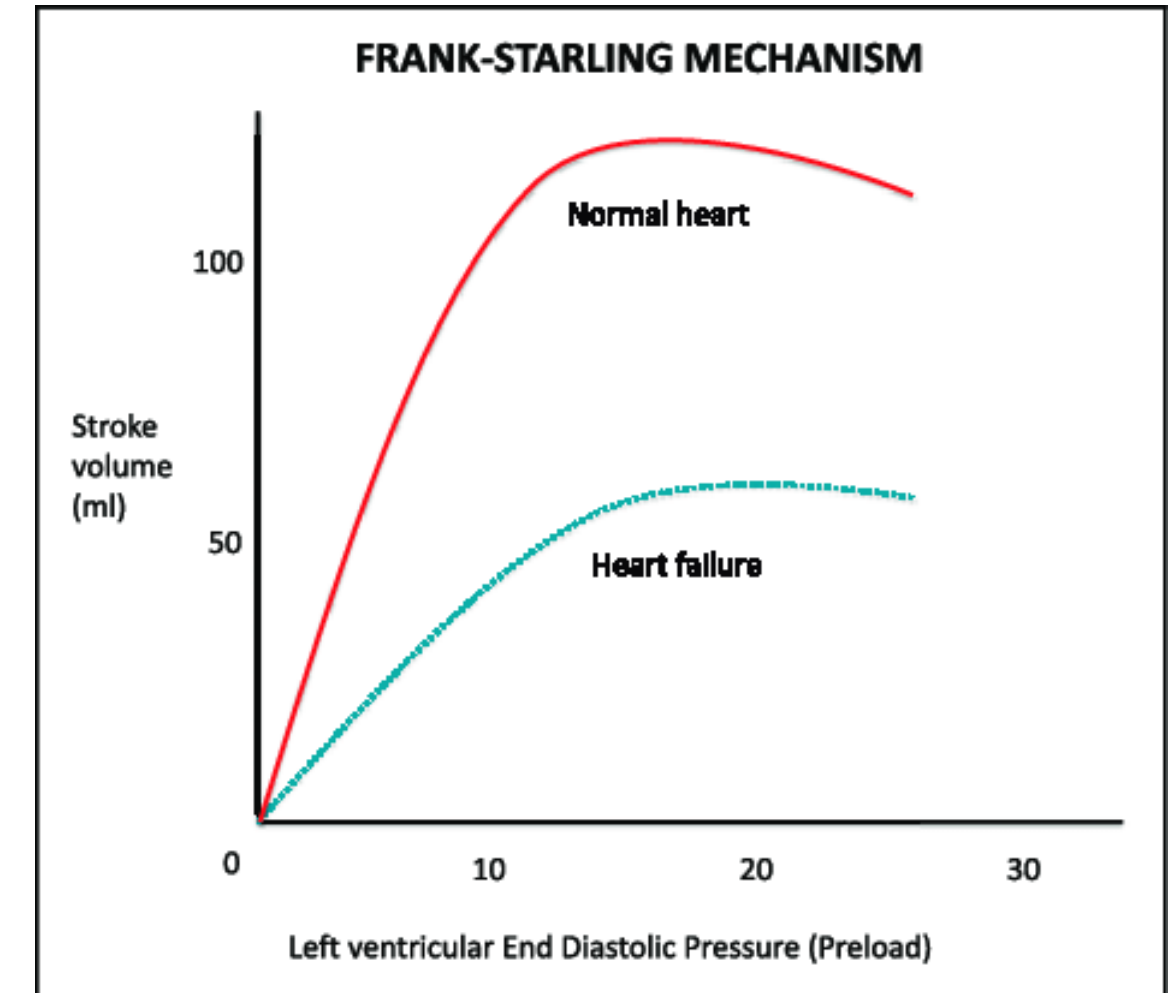
INOTROPIC AGENTS



- Sympathetic NS
- Adrenergic agonists
- Cardiac glycosides (digoxin)
- Angiotensin II
- Low Na^+
- High Heart rate

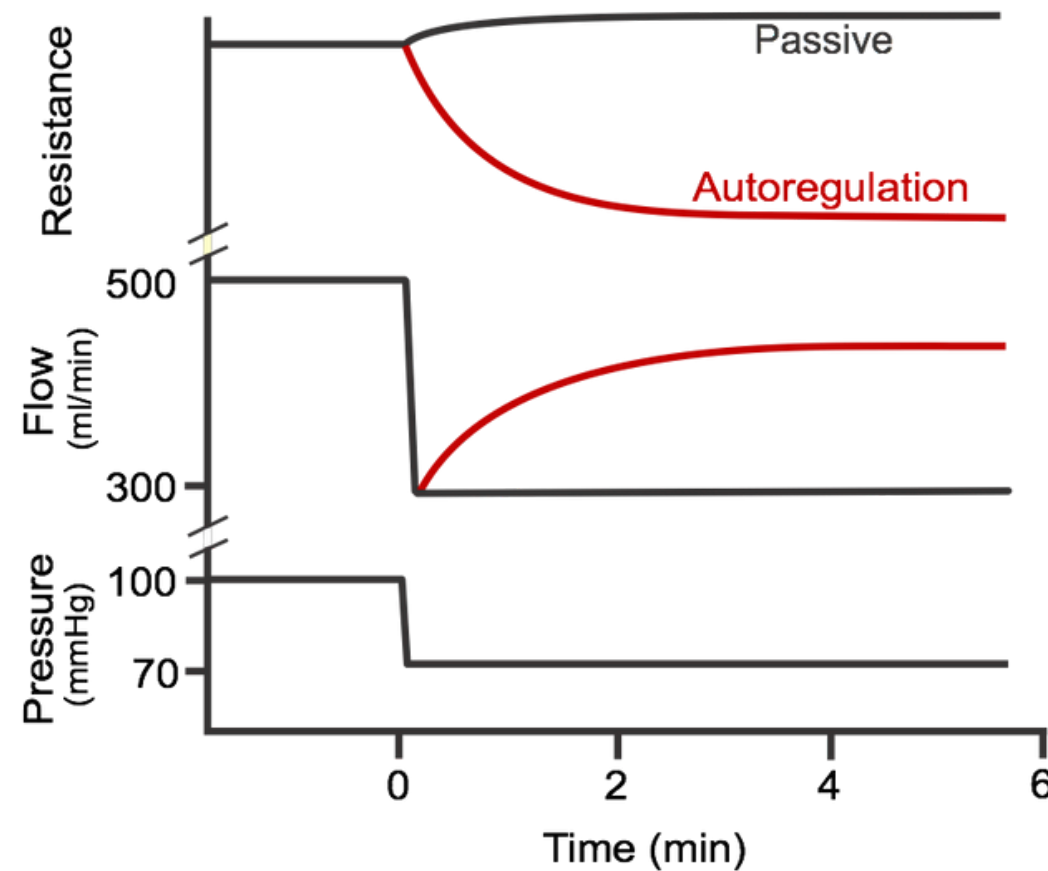


- Parasympathetic NS (atria)
- Beta blockers
- Low Ca^{2+} (channel blockers)
- Some antiarrhythmics
- High Na^+
- Low Heart rate

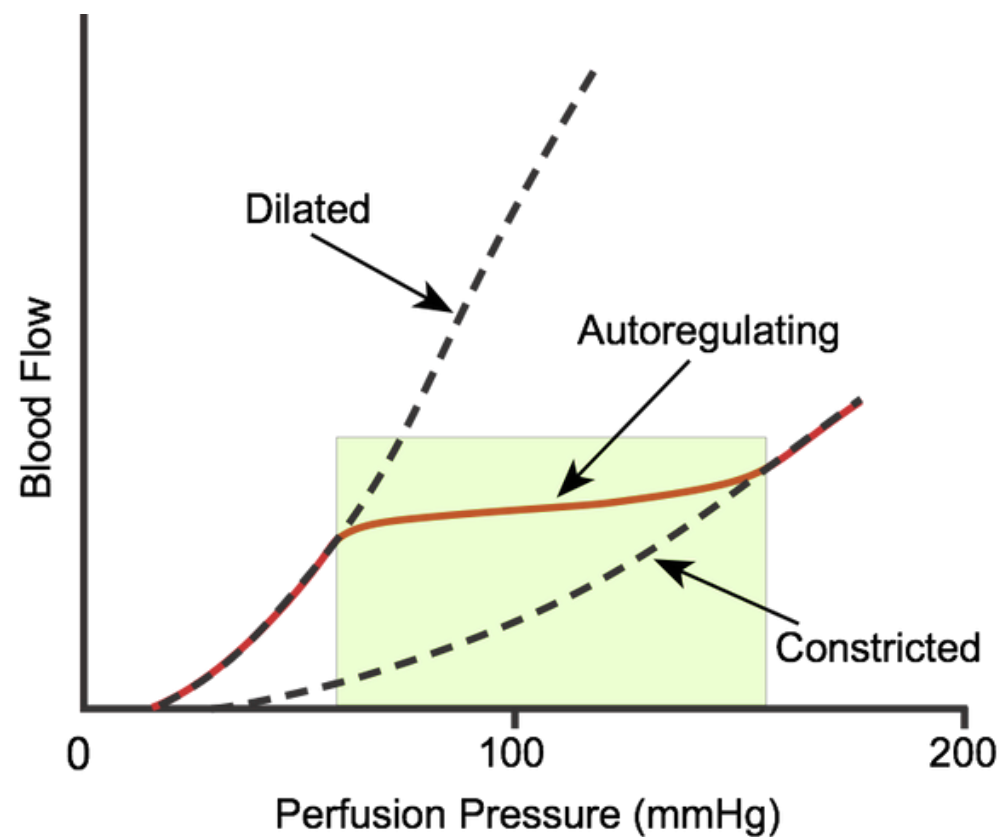


- Under resting conditions, the heart rate is approximately 72 beats/min, and the stroke volume is approximately 70 mL/beat. Consequently, **the cardiac output is slightly more than 5 L/min.**
- During exercise, the heart rate in a nonathlete can increase to 190 bpm, and the stroke volume can increase to 115 mL/beat. Therefore, the **cardiac output increases to approximately 22 L/min:**

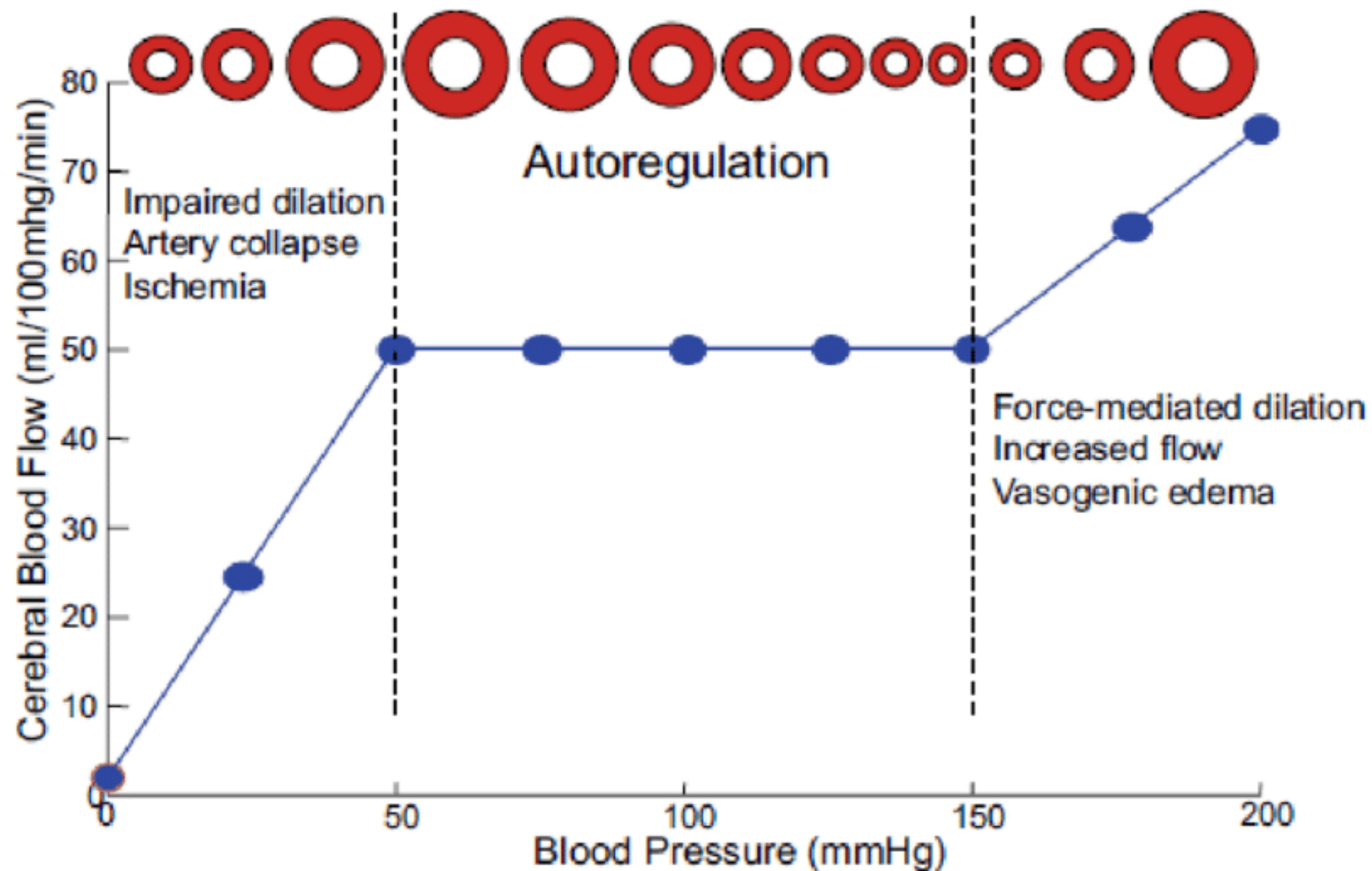
Intrinsic control in vascular function (autoregulation)



- the body's intrinsic ability to maintain constant blood flow to an organ despite fluctuations in blood pressure.
- This is achieved through the active dilation or constriction of blood vessels, primarily **arterioles**, in response to changes in pressure, metabolites, or shear stress.
- This process ensures critical organs like the brain and heart receive a stable supply of blood and oxygen, but its effectiveness varies among different organs.



Cerebral autoregulation and the vascular response to blood pressure change.



Active hyperemia

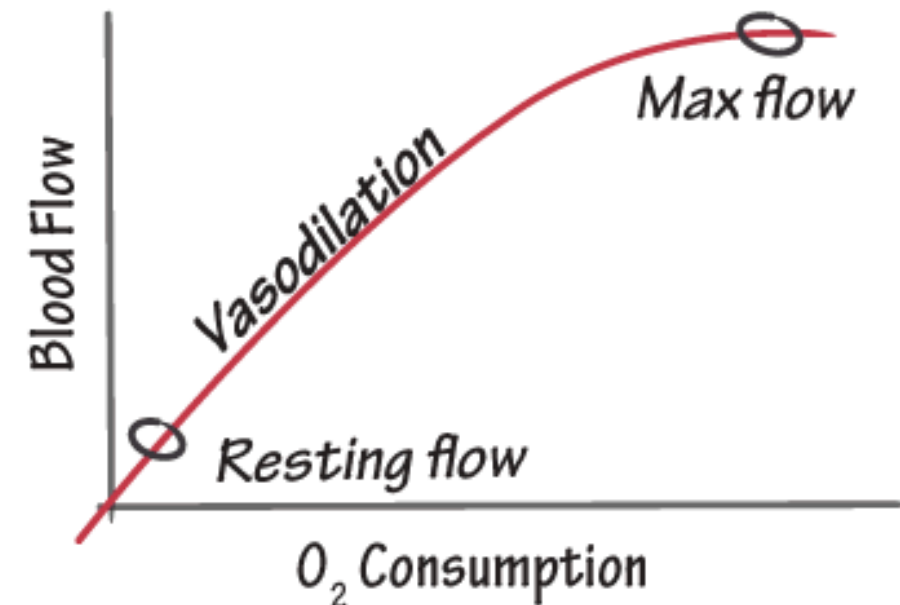
Reflects positive correlation b/w blood flow and tissue metabolic requirements.

Skeletal muscle

- ↑ Skeletal muscle activity;
- ↑ O_2 consumption; Vasodilation;
- ↑ Blood flow and oxygen supply.

Fits the:

■ Metabolic hypothesis



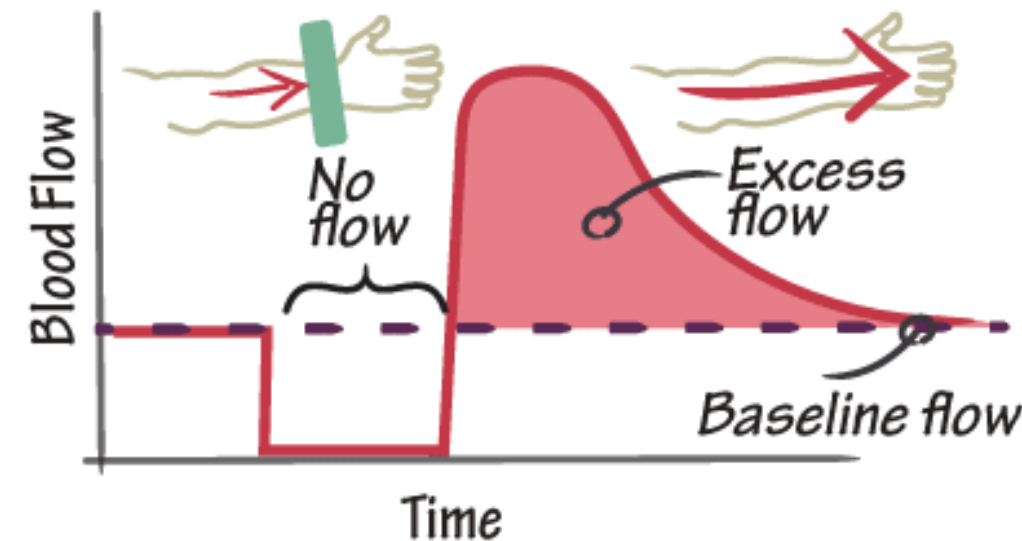
Re-active hyperemia

Response to period of decreased blood flow.

Obstruct blood flow
 O_2 debt accumulates

↑↑ Enable blood flow
 Blood flow until O_2 debt is repaid.

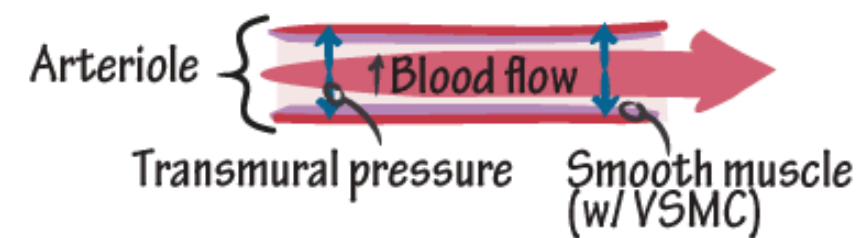
Fits the:
 ■ Metabolic hypothesis



HYPOTHESES

Myogenic

Maintain vascular wall tone.



1. ↑ Blood flow ↑ Transmural pressure.
2. Stretch receptors in VSMC initiate vasoconstriction.



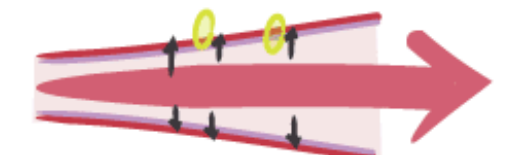
3. Contraction maintains vascular tone and ↓ blood flow.

Metabolic

Match oxygen supply and demand.



1. ↑ Tissue activity;
 ↑ Vasodilator metabolites.



2. Vasodilator metabolites induce dilation.
3. Blood flow and Oxygen supply ↑.

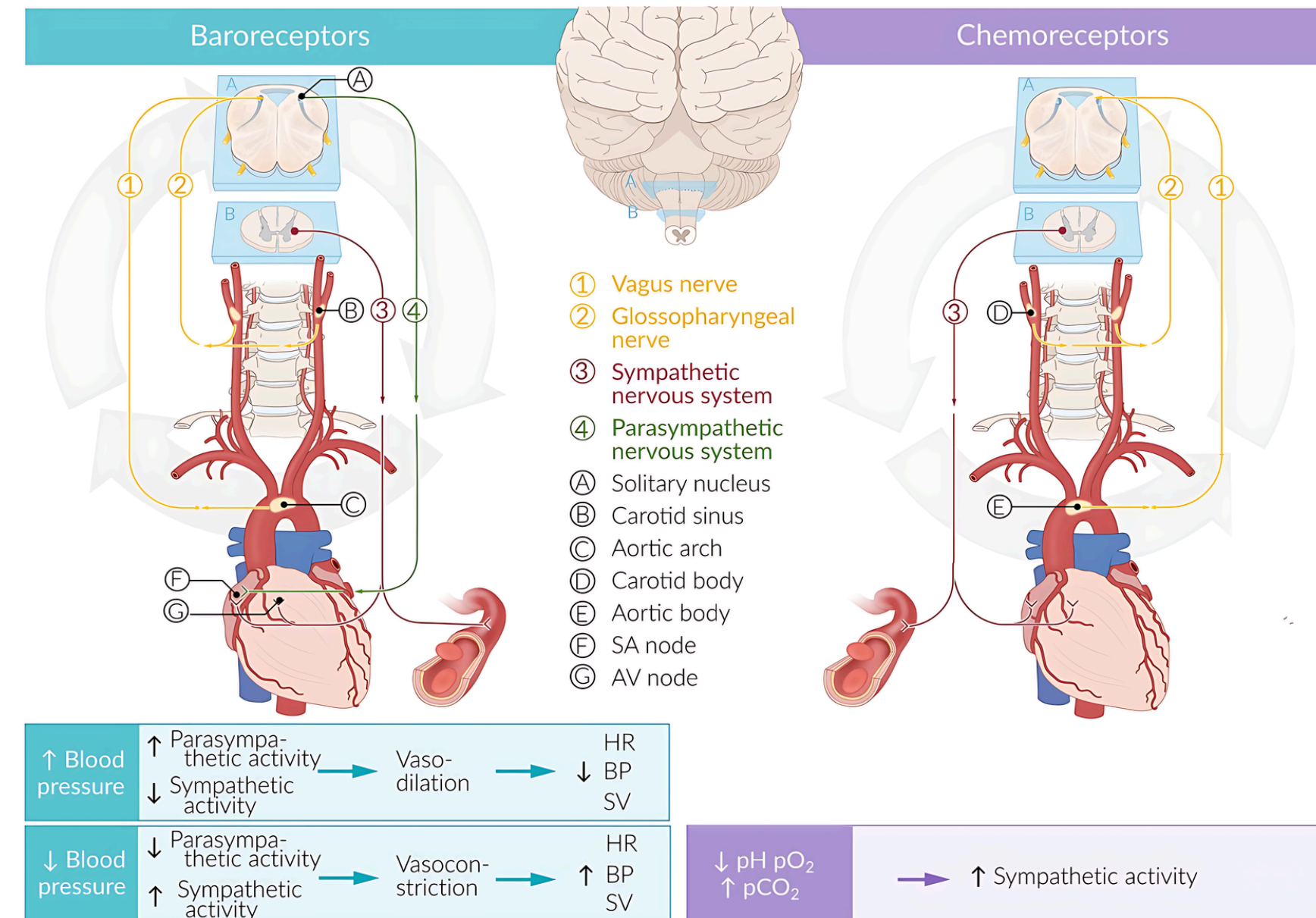
Short term mechanism

1. Baroreceptor reflex:

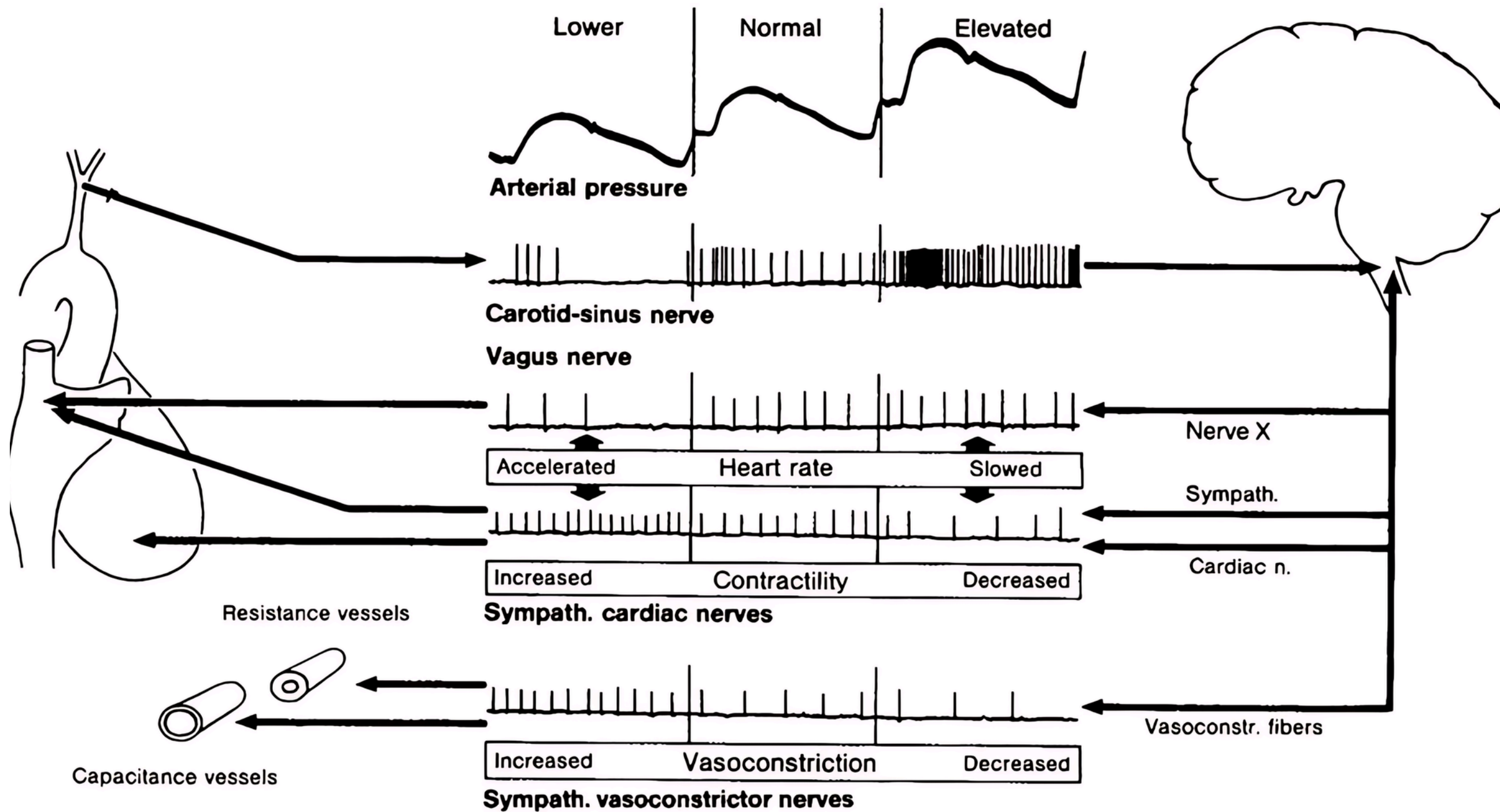
- The afferent component is composed of **pressure-sensitive receptors** associated with two major blood vessels, the aortic arch (vagus nerve) and the carotid sinus (glossopharyngeal nerve), and their projections into the CNS.
- These baroreceptor afferents all project into the CNS and terminate in the dorsal medial medulla in a distinct region of the nucleus tractus solitarius (NTS).

Any change in your body's demand for blood can trigger your baroreceptor reflex:

- **Change your body position**, such as when you stand.
- See or experience something that **frightens**.
- Switch from walking to **running**.

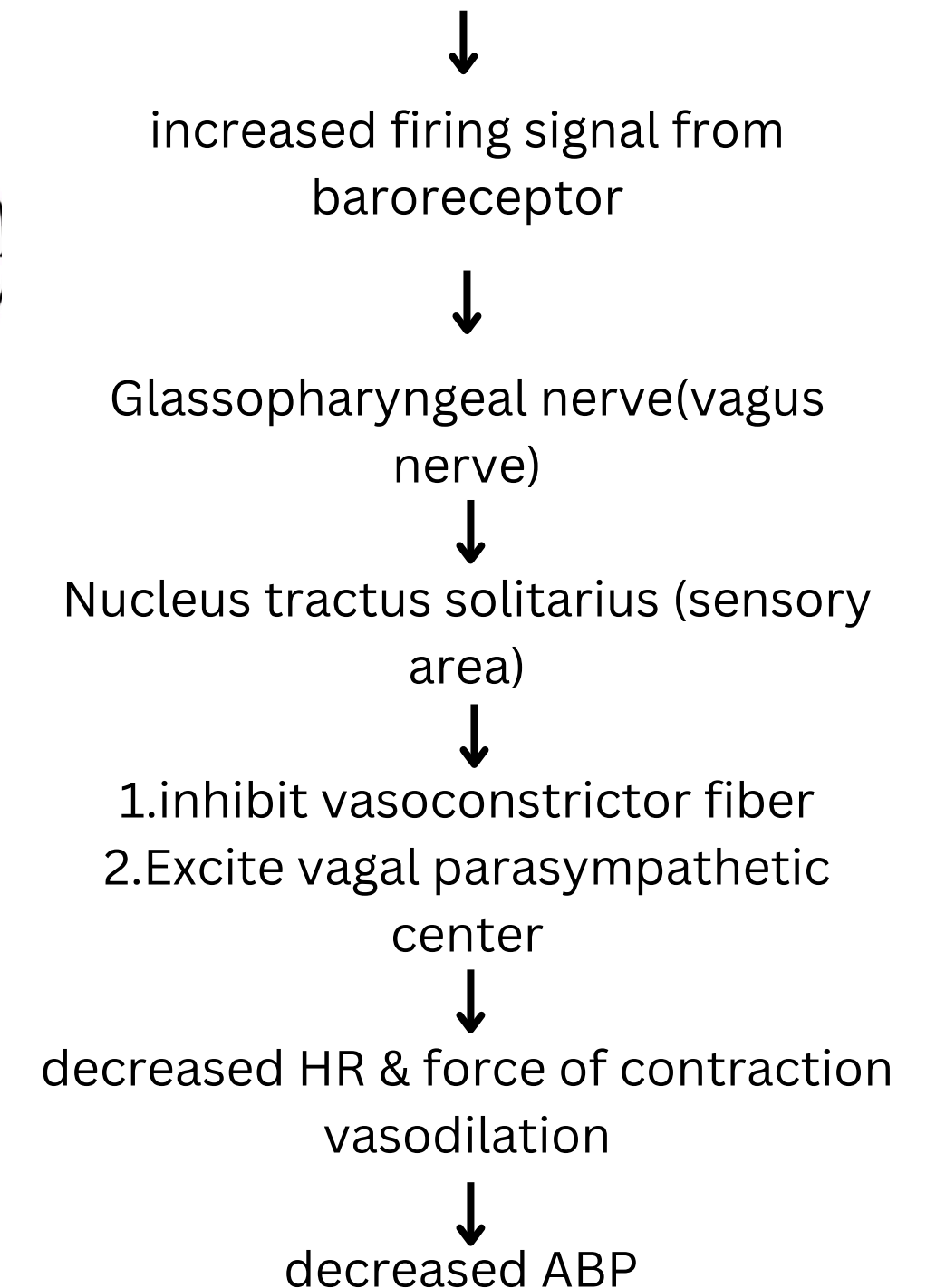


Response of baroreceptor to pressure change

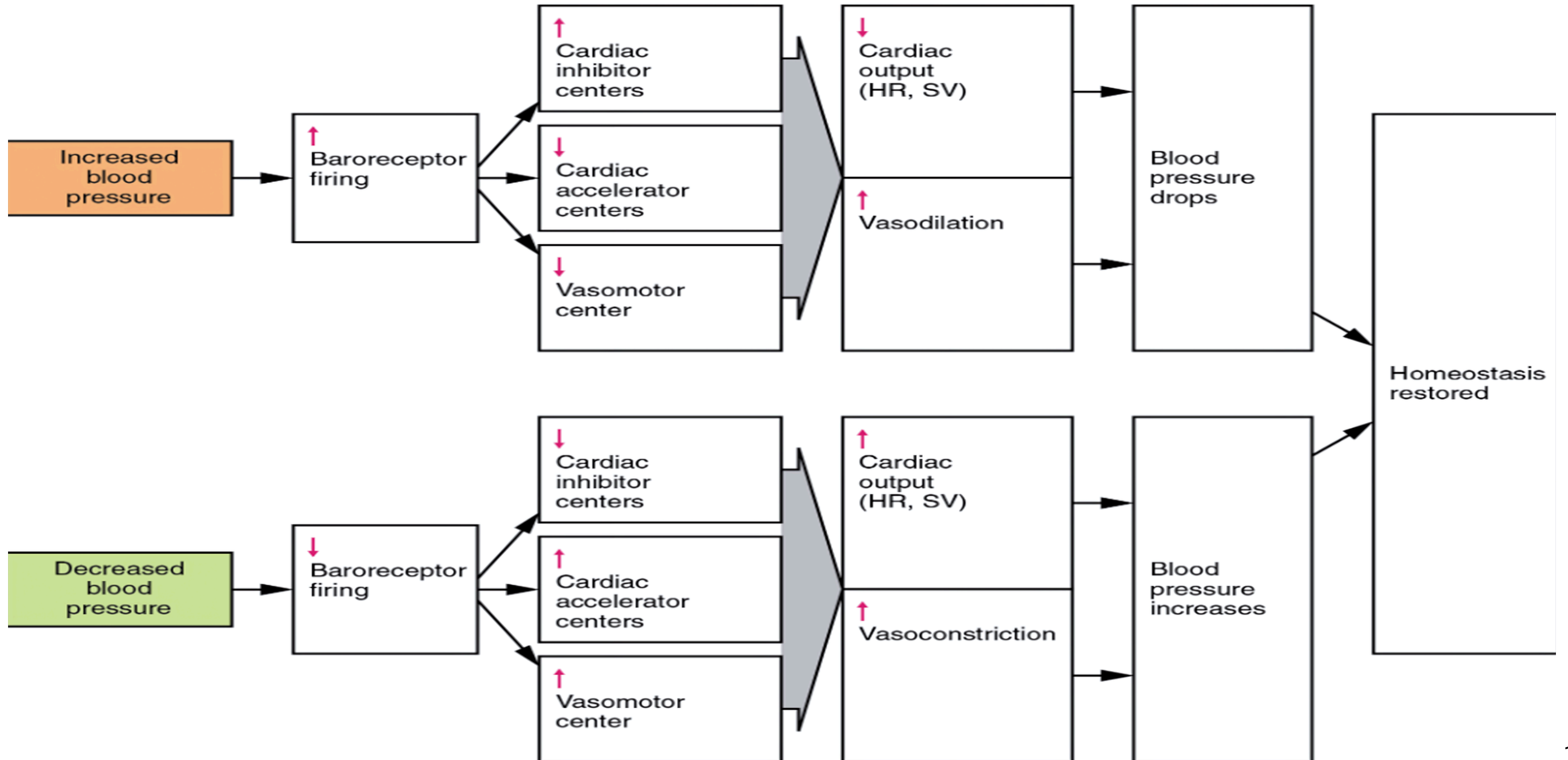


(Cheryl M Heesch, 2000)

ABP > 60 mm Hg (carotid sinus)
ABP > 90 mm Hg (aortic arch)



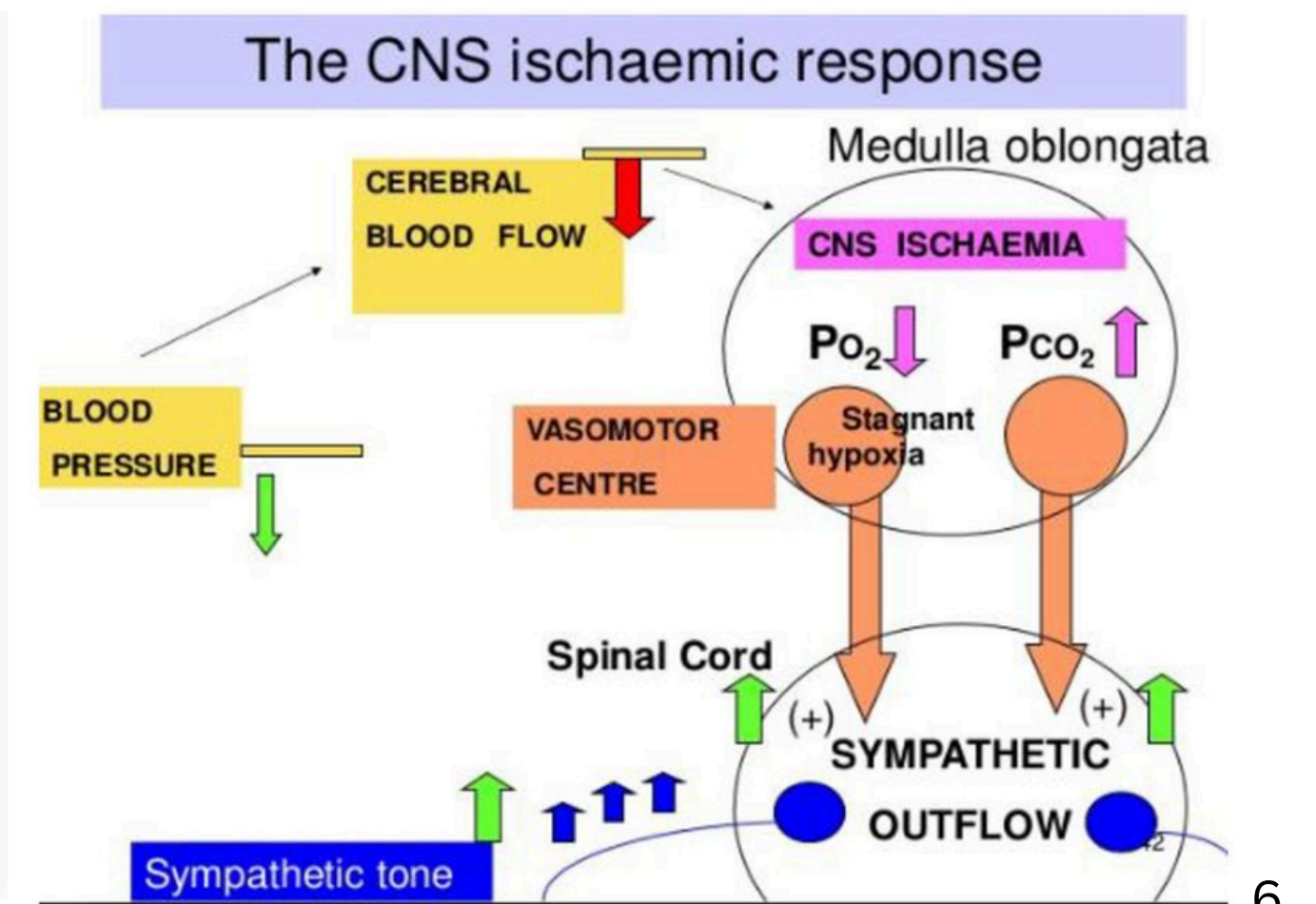
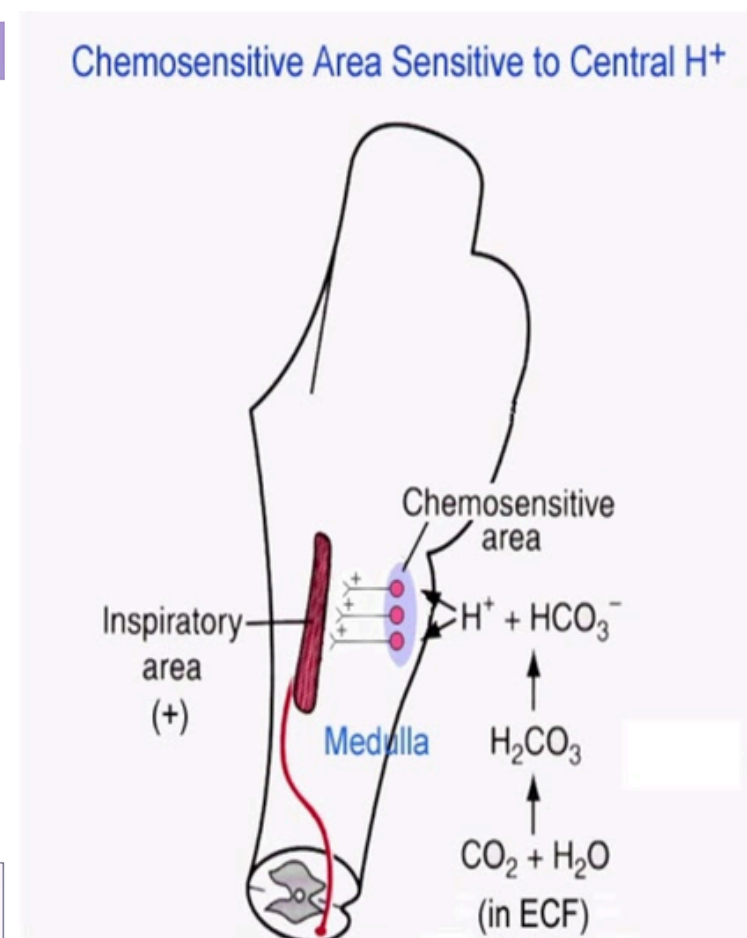
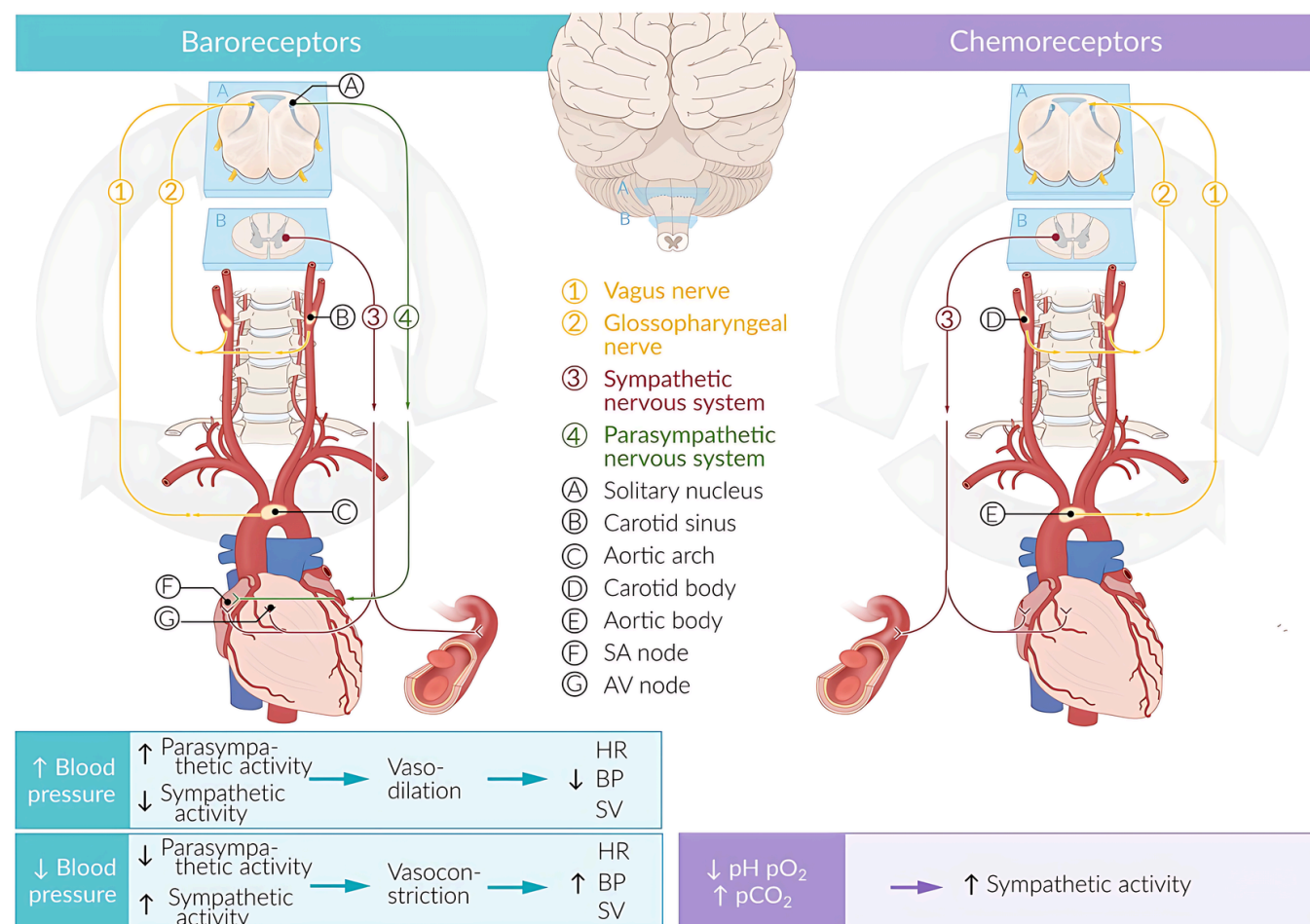
Response of baroreceptor to pressure change



Short term mechanism

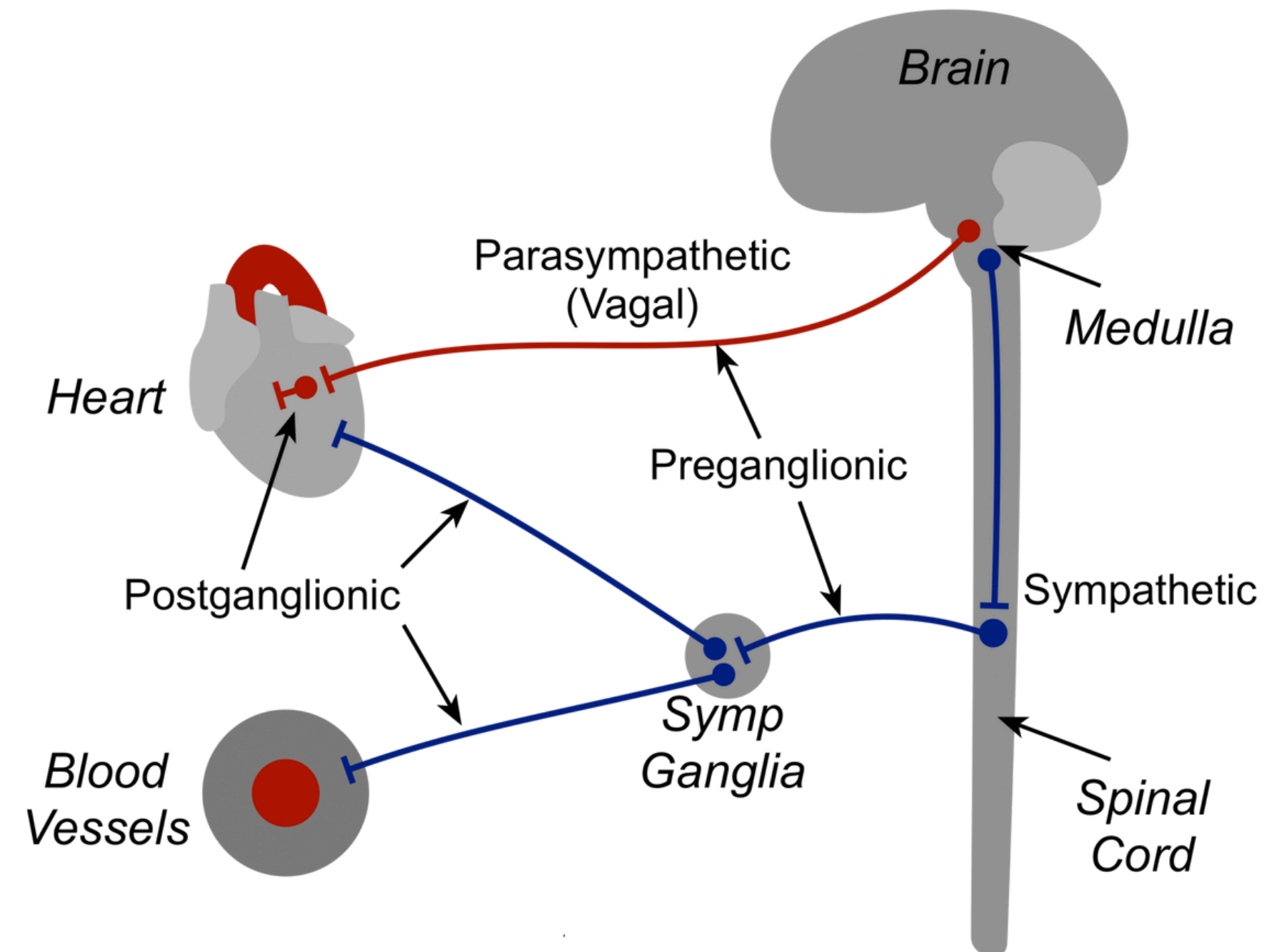
2. Chemoreceptor reflex:

- The chemoreflexes are important modulators of sympathetic activation.
- The peripheral chemoreceptors located in the carotid bodies respond primarily to hypoxaemia.
- **Central chemoreceptors** located in the region of the brainstem (medulla) respond to **hypercapnia**.
- Activation of either the **hypoxic or hypercapnic chemoreflex elicits both hyperventilation and sympathetic activation**.
- The peripheral and central chemoreflexes have powerful effects on sympathetic activity in both health and disease and may contribute importantly to disease pathophysiology, particularly in conditions such as hypertension, obstructive sleep apnea and heart failure.



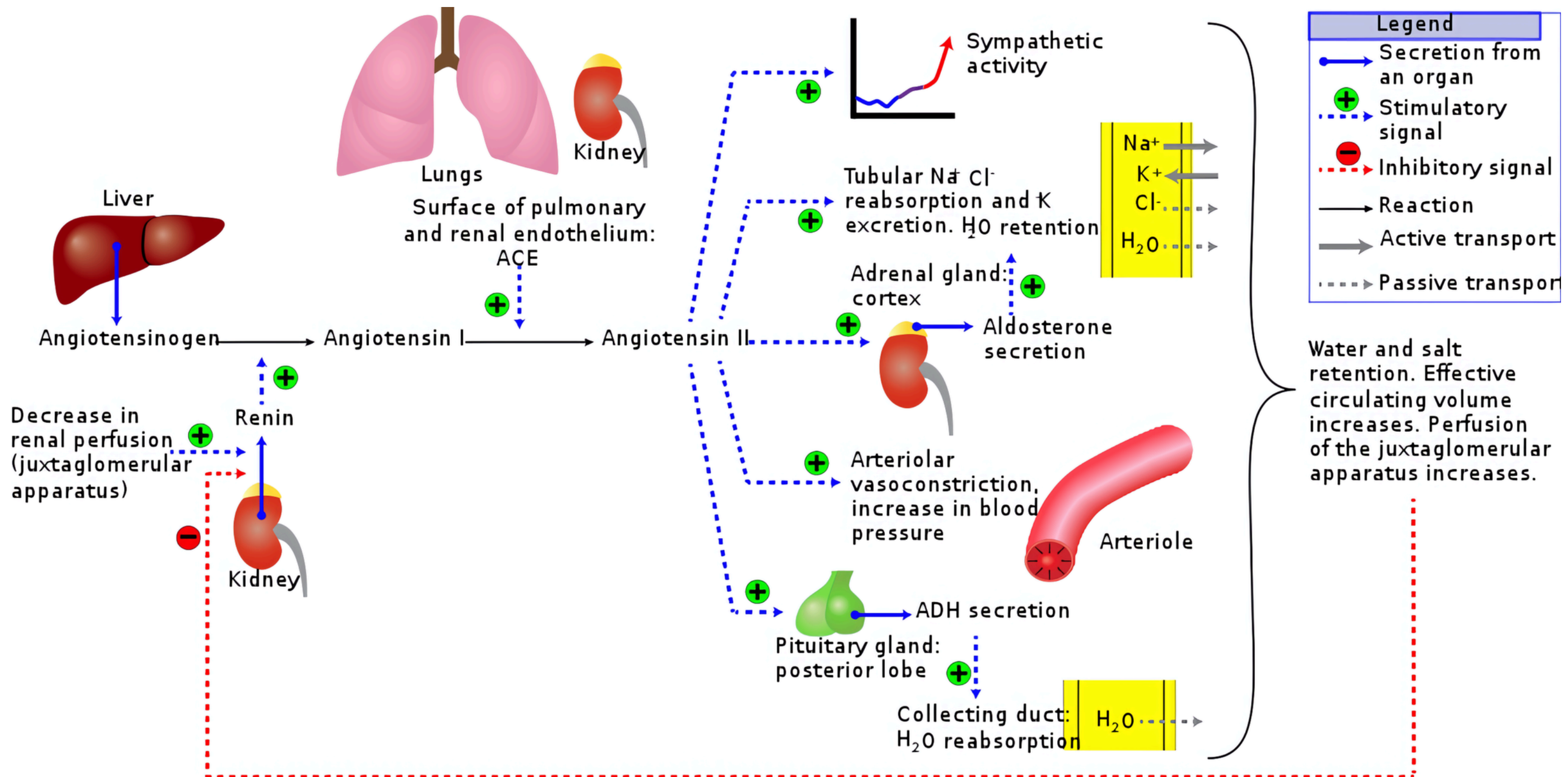
Summary of **autonomic reflexes** affecting heart & circulation

Reflex	Receptor type	Receptor area	Stimulus	Sym activity	Parasym activity
Arterial baroreceptor	Mechano receptor	Internal carotids and aortic arch	↑ASP	↓	↑
Bainbridge	Mechano receptor	Veno-atrial junctions	↑venous return	↑	↓
Cardiac	Mechano receptor	Atria & Ventricles	↑chamber pressure	↓	↑
Peripheral chemoreceptor	Chemoreceptor	Carotid and aortic bodies	↓PO ₂ , ↑PCO ₂ , acidosis	↑	↓
Central chemoreceptor	Chemoreceptor	Medulla oblongata	↑PCO ₂ , acidosis	↑	↓
Cushing reflex	Chemoreceptor	Brain	↑Intracranial pressure	↑	↑
Cerebral ischemia	Chemoreceptor	Brain	Ischemia	↑	↓
Bezold-Jarisch reflex	Chemoreceptor	Ventricles & coronary artery	Chemical, ischemia	↓	↑
Pain	Nociceptor	Various	Pain	↑	↓
Deep pain	Nociceptor	Various	Pain	↓	↑
Vasovagal reflex	Various	Various	Strong emotion, pain	↓	↑
Pulmonary stretch reflex	Proprioceptor	Airways, respiratory muscles	Lung inflation	↓	↓
Muscle & joint reflex	Proprioceptor	Muscle	Muscle movement	↑	↓
Diving reflex	Thermoreceptor	Face	Water submersion	↑	↑
Temperature reflex	Thermoreceptor	Skin, Hypothalamus	↑ or ↓ temperature	↑ or ↓	Insignificant



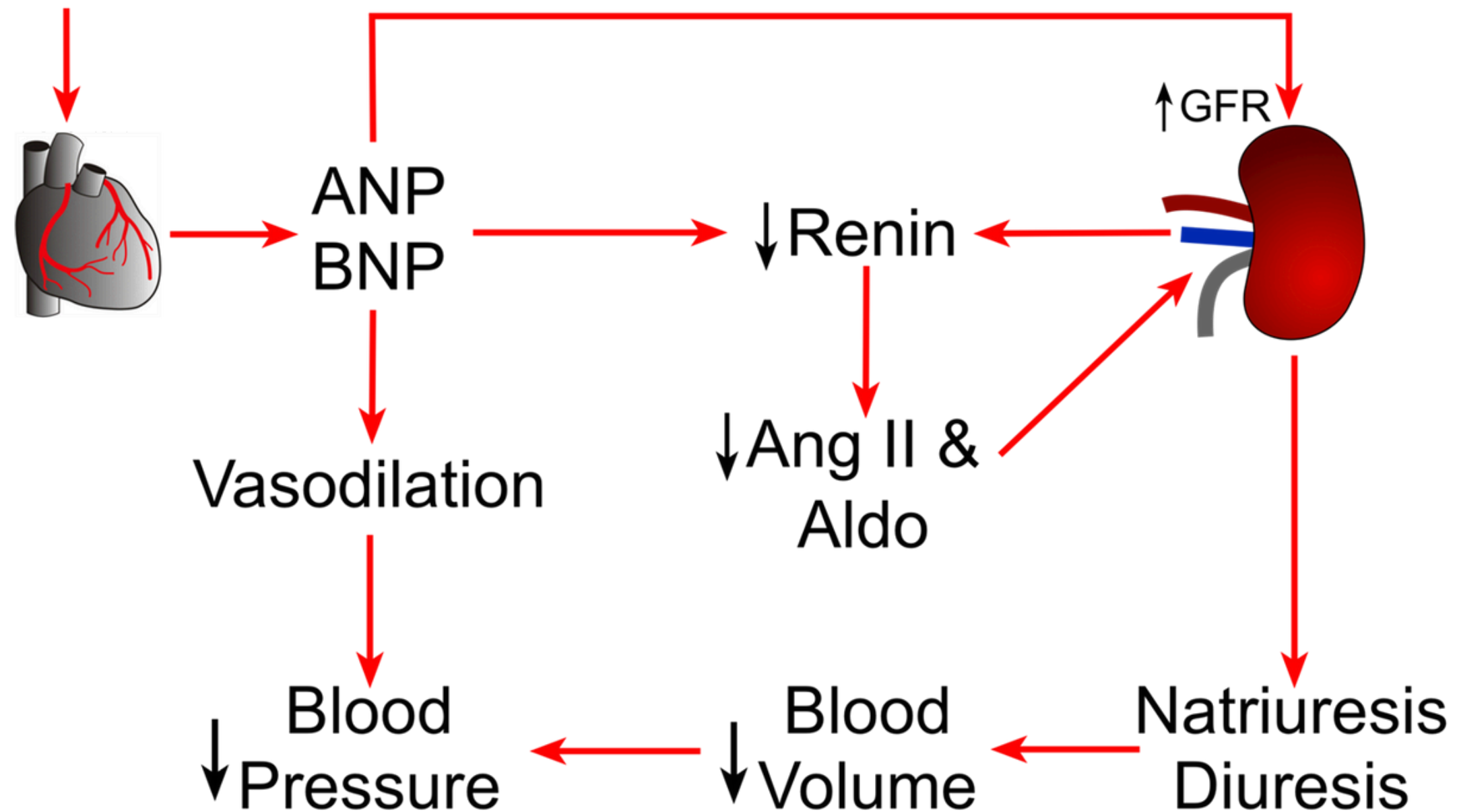
Long term mechanism

1. Renin-angiotensin-aldosterone system (RAAS)



2. Atrial natriuretic peptide (ANP)

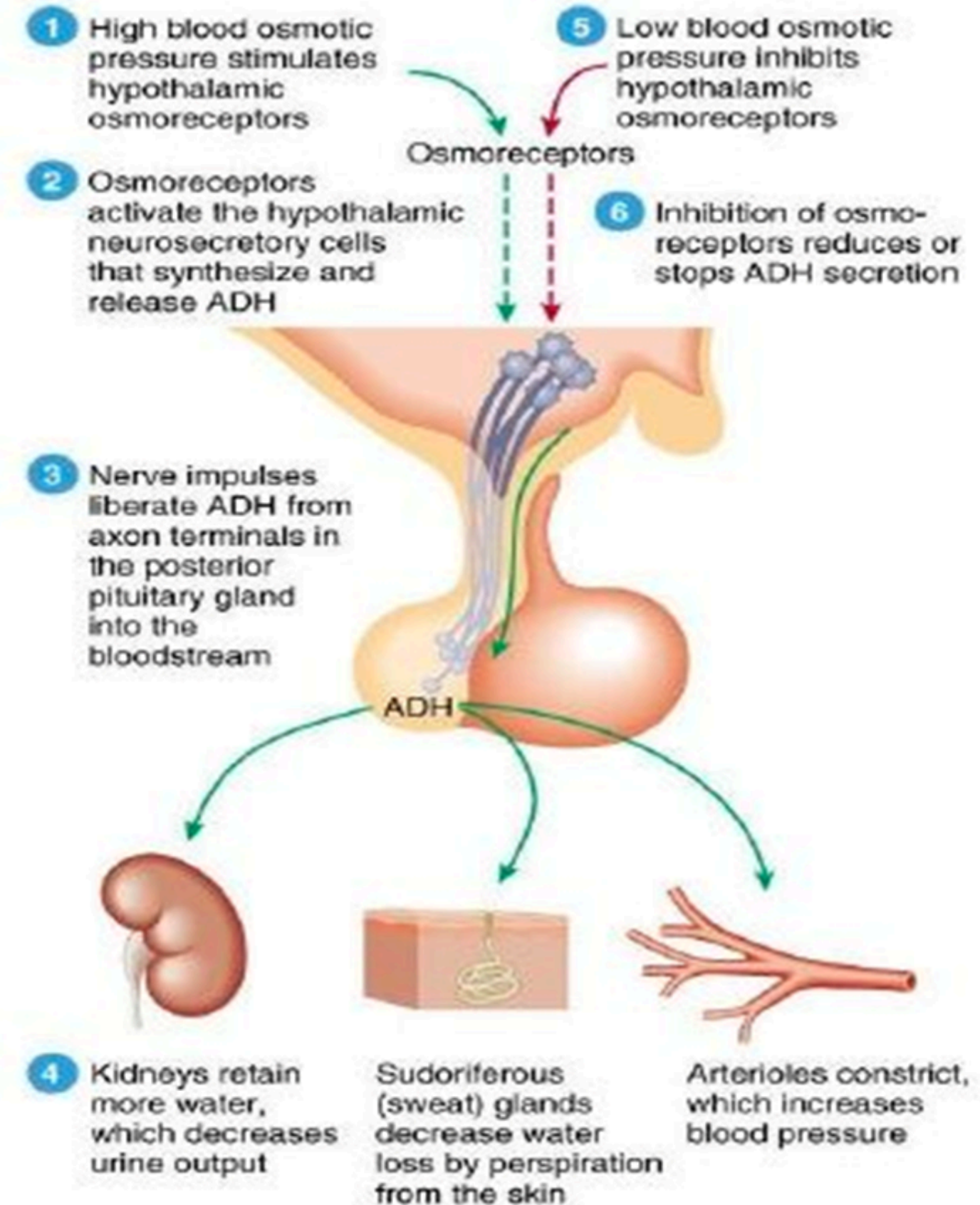
Cardiac distension
Sympathetic stimulation
Angiotensin II

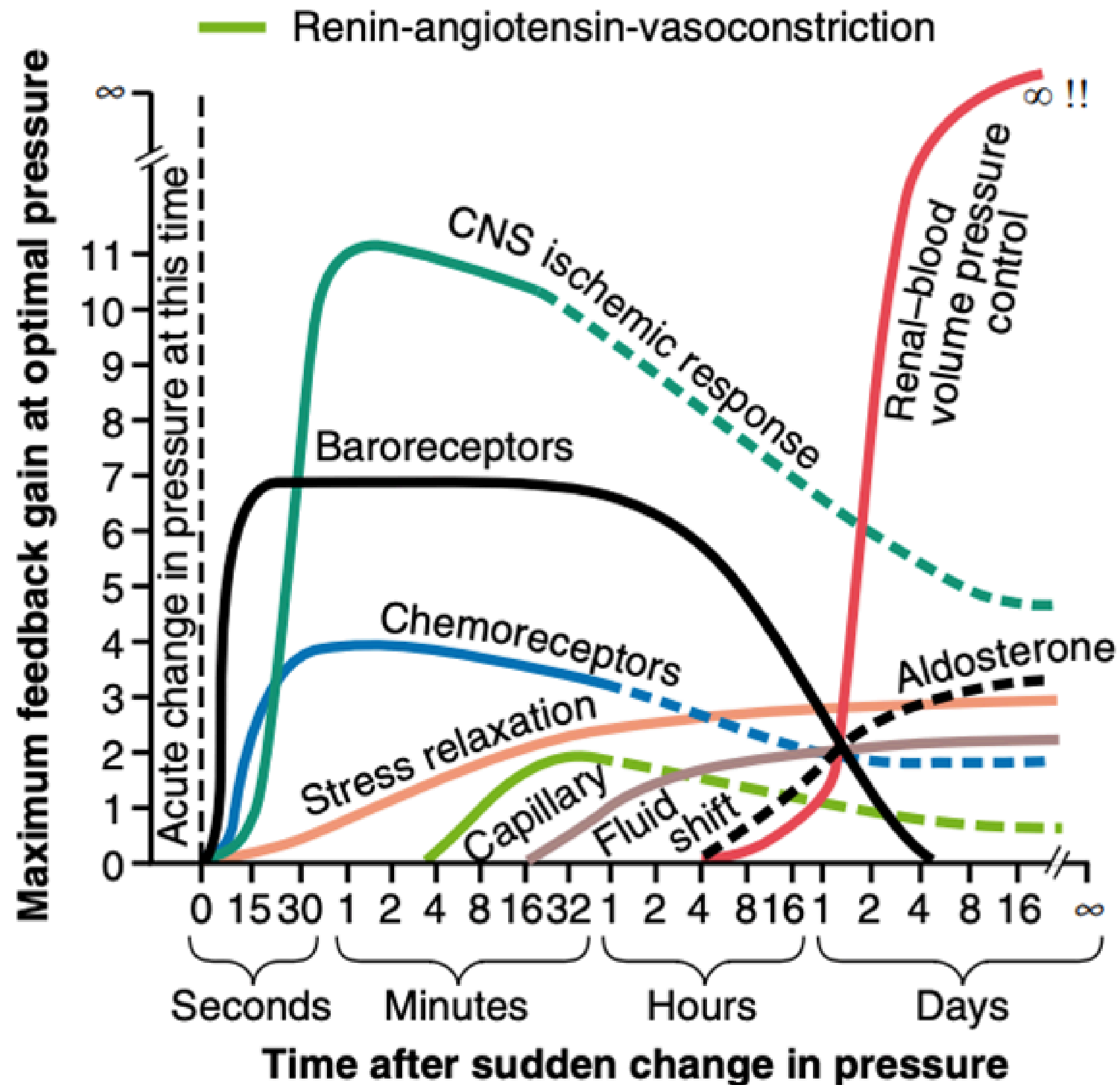


Long term mechanism

3. Antidiuretic hormone (ADH)

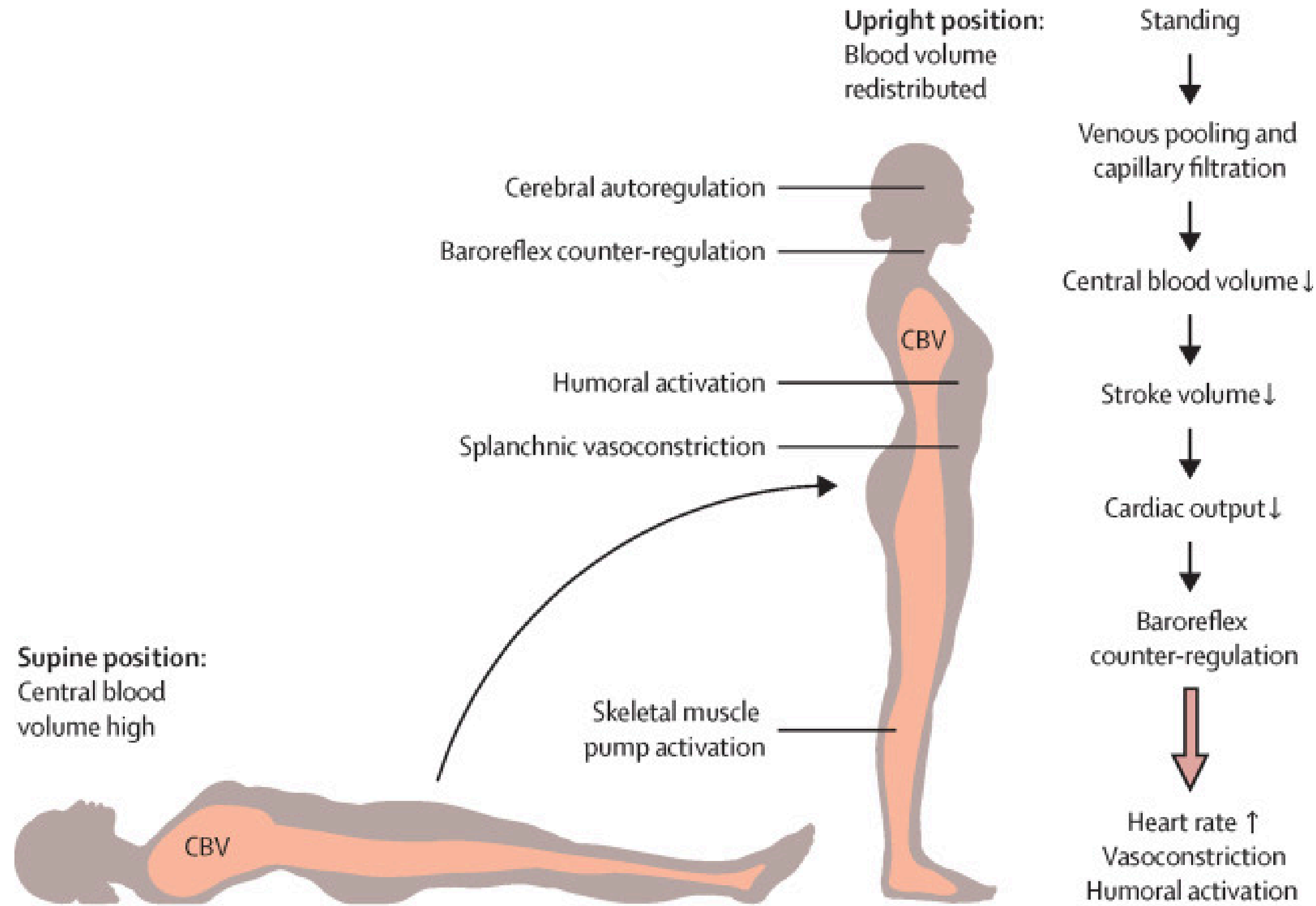
- Known as **vasopressin**
- Functions
 - decrease urine production
 - decrease sweating
 - increase BP
- Dehydration
 - ADH released
- Overhydration
 - ADH inhibited



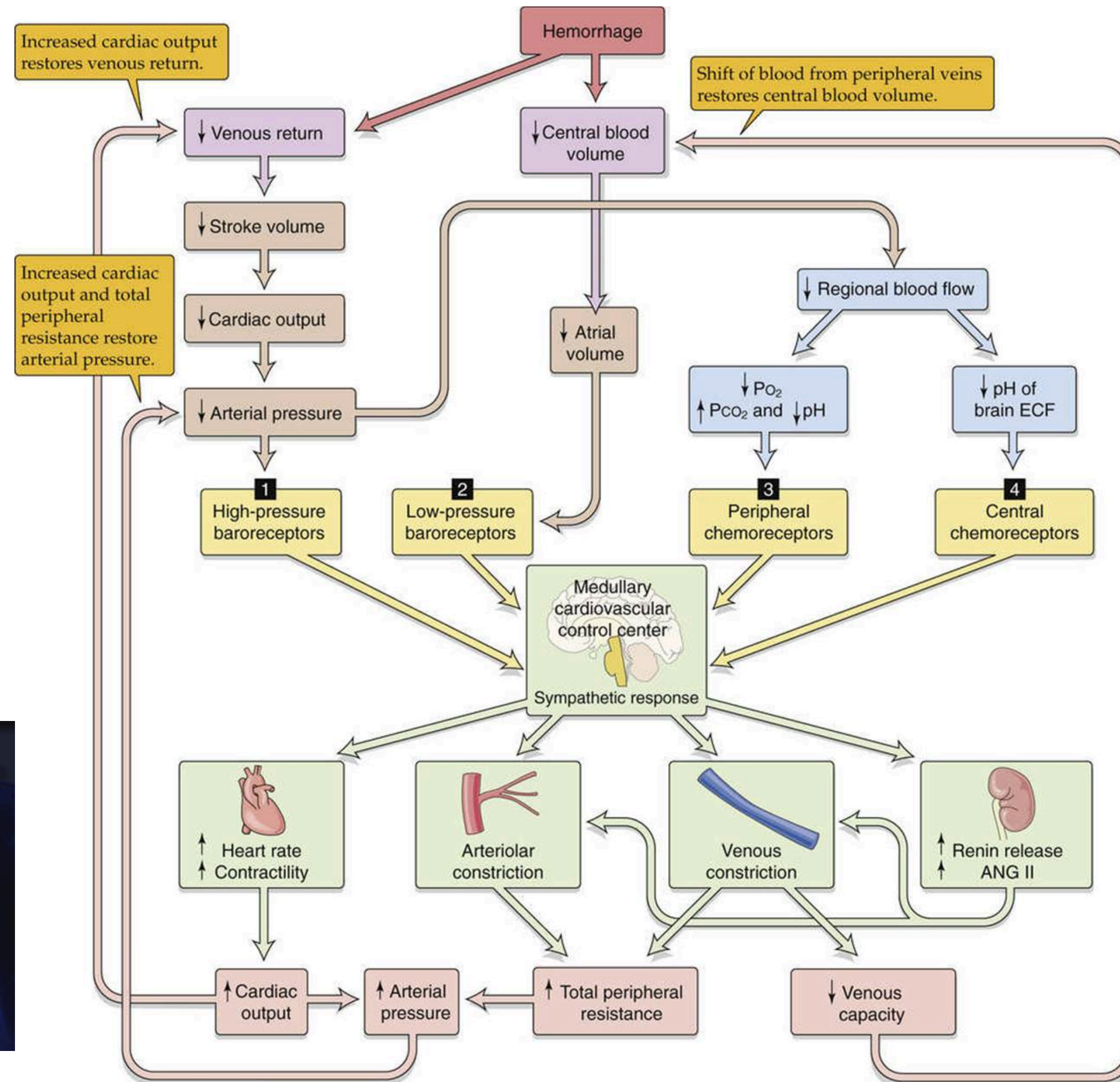


- The **systemic regulatory** mechanisms that synergize with the local mechanisms to adjust vascular responses throughout the body

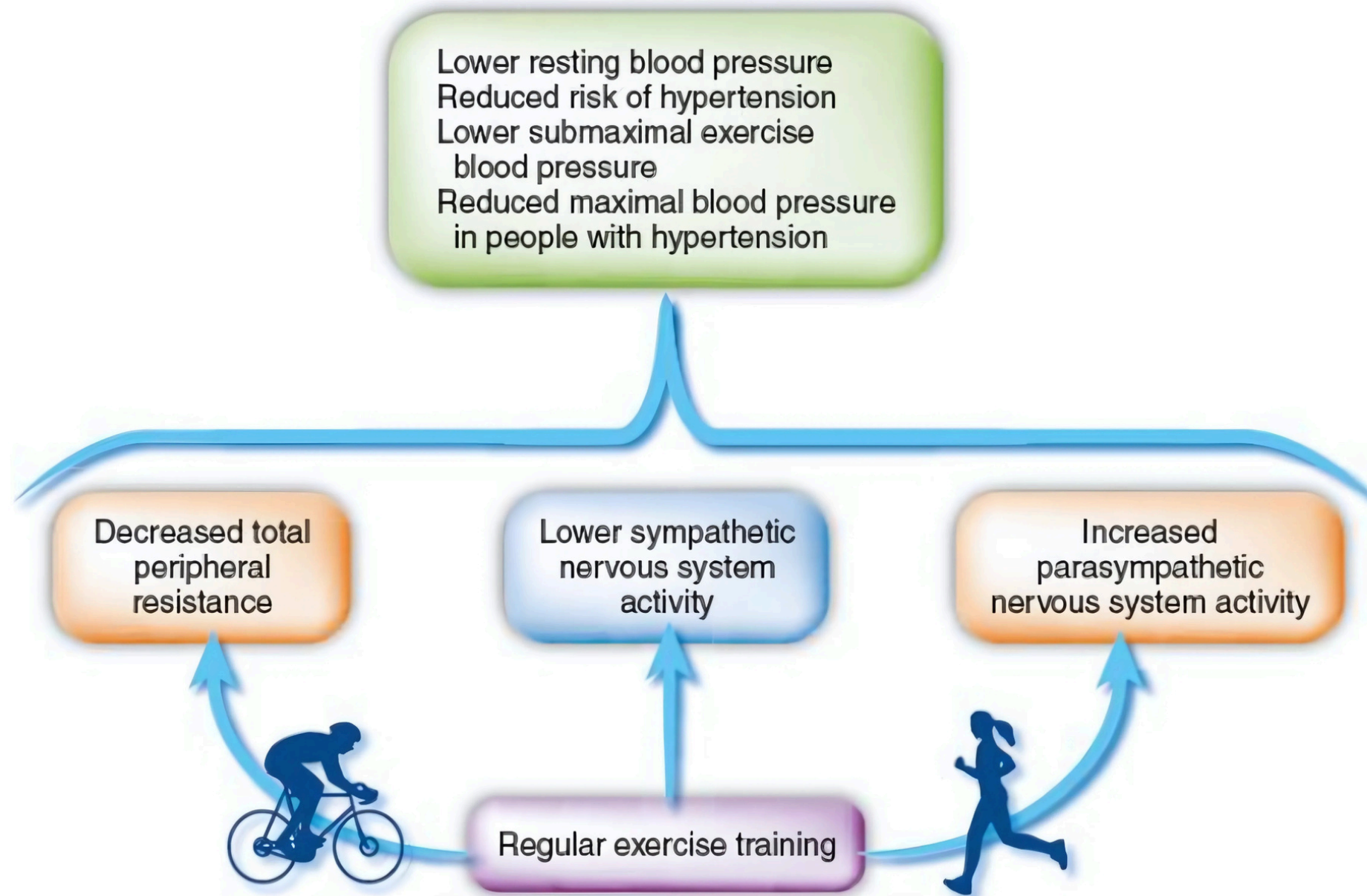
Postural changes



Blood loss



The response of CVS to exercise



Source: John P. Porcari, Cedric X. Bryant, Fabio Comana: Exercise physiology
www.FADavisPTCollection.com
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